

## Project based on creation of "Employee Attrition Dashboard"

### ✔ Data Gathering:

- ✔ In PBIDW goto "Get Data" and load the csv file named "HR\_Analytics\_Dataset.csv".
- ✔ It contains 1480 rows and 36 columns.
- ✔ Dataset Attribute Descriptions:

| Column Name             | Description                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------|
| EmpID                   | <b>Employee ID</b> – Unique identifier for each employee. E.g., RM297, RM302                       |
| Age                     | Employee's age in years. E.g., 18, 19                                                              |
| Attrition               | Indicates whether the employee has left (Yes) or is still with the company (No).                   |
| BusinessTravel          | Frequency of travel required: Travel_Rarely, Travel_Frequently, Non-Travel                         |
| DailyRate               | Daily earning or billing rate of the employee. E.g., 230, 1306                                     |
| Department              | Department the employee works in: Sales, Research & Development, etc.                              |
| DistanceFromHome        | Distance from employee's home to workplace (in kms or miles). E.g., 3, 10, 22                      |
| Education               | Education level as numeric category:<br>1=Below College, 2=College, 3=Bachelor, 4=Master, 5=Doctor |
| EducationField          | Specialization in education. E.g., Life Sciences, Medical, Marketing                               |
| EmployeeCount           | Always 1 – used as a placeholder (can be dropped).                                                 |
| EmployeeNumber          | System-generated unique number for employee. E.g., 405, 614                                        |
| EnvironmentSatisfaction | Satisfaction with work environment: 1 (Low) to 4 (Very High)                                       |
| Gender                  | Employee gender: Male or Female                                                                    |
| HourlyRate              | Hourly rate of pay. E.g., 54, 69, 80                                                               |
| JobInvolvement          | Level of involvement in job: 1 (Low) to 4 (Very High)                                              |
| JobLevel                | Job seniority level: 1 (Entry) to 5 (Executive)                                                    |
| JobRole                 | Designation or role: Sales Representative, Laboratory Technician, etc.                             |
| JobSatisfaction         | Satisfaction with job: 1 (Low) to 4 (Very High)                                                    |
| MaritalStatus           | Marital status: Single, Married, or Divorced                                                       |
| MonthlyIncome           | Monthly base salary (net/gross). E.g., 1420, 1569                                                  |

| Column Name              | Description                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------|
| MonthlyRate              | Monthly rate billed to client or internal standard rate. (Earnings with incentives, bonus, or overtime included) |
| NumCompaniesWorked       | Total number of previous companies worked at.                                                                    |
| Over18                   | Indicates employee is over 18: Always Y (can be dropped).                                                        |
| OverTime                 | Does the employee work overtime: Yes or No                                                                       |
| PercentSalaryHike        | Recent percentage salary hike (increment).                                                                       |
| PerformanceRating        | Rating in most recent performance review: 1 (Low) to 4 (Excellent)                                               |
| RelationshipSatisfaction | Satisfaction with colleagues: 1 (Low) to 4 (Very High)                                                           |
| StandardHours            | Always 80 (default full-time hours in many orgs) – can be dropped                                                |
| StockOptionLevel         | Stock options given: 0 (None) to 3 (High)                                                                        |
| TotalWorkingYears        | Total years of work experience                                                                                   |
| TrainingTimesLastYear    | Number of training programs attended last year                                                                   |
| WorkLifeBalance          | Rating of balance between work and personal life: 1 (Bad) to 4 (Excellent)                                       |
| YearsAtCompany           | Total years worked in this company                                                                               |
| YearsInCurrentRole       | Years spent in current job role                                                                                  |
| YearsSinceLastPromotion  | Years since the employee was last promoted                                                                       |
| YearsWithCurrManager     | Years working with the current manager                                                                           |

---

### Data Cleaning:

- ☒ In PQEW, under View tab enable "Column quality". Check null values in all the columns and delete the column "YearsWithCurrManager" as it is containing 4% null values and also not required column. (Optional)
- ☒ Do the Group by on "EmpID" with row count to check for duplicate rows. And found 10 duplicate rows on "EmpID". Delete this step.
- ☒ In the main query table select all the columns and go for "Remove Duplicates". Now after remove duplicates 1480 rows will be reduced to 1473 rows.
- ☒ Check for misspelled entries. Under column "BusinessTravel" there we shall replace "TravelRarely" with "Travel\_Rarely".
- ☒ Select all columns and goto "Transform" and "Detect Data Type" to assign right datatypes to all the columns. Now click "Close & Apply" and return to PBIDW.

- ☒ In PBIDW insert following columns:
    - ☒ Insert column **AttritionFlag** against formula **AttritionFlag = IF(HR\_Analytics\_Dataset[Attrition] = "Yes", 1, 0)**
    - ☒ Insert column **SalarySlab** against formula **SalarySlab = if(HR\_Analytics\_Dataset[MonthlyIncome] <= 5000, "Upto 5K", if(HR\_Analytics\_Dataset[MonthlyIncome] <= 10000, "5K+ to 10K", if(HR\_Analytics\_Dataset[MonthlyIncome] <= 15000, "10K+ to 15K", "15K+")))**
    - ☒ Insert column **AgeGroup** against formula **AgeGroup = if(HR\_Analytics\_Dataset[Age] <= 25, "18-25", if(HR\_Analytics\_Dataset[Age] <= 35, "26-35", if(HR\_Analytics\_Dataset[Age] <= 45, "36-45", if(HR\_Analytics\_Dataset[Age] <= 55, "46-55", "55+"))))**
    - ☒ Create a measure variable **AttritionRate** against this DAX formula **AttritionRate = DIVIDE(SUM(HR\_Analytics\_Dataset[AttritionFlag]), COUNT(HR\_Analytics\_Dataset[EmpID]))**. Make it of % type and with one decimal digit.
    - ☒ Otherwise we can create the same measure variable **AttritionRate** against the DAX formula **AttritionRate = DIVIDE(CALCULATE(COUNT(HR\_Analytics\_Dataset[EmpID]), HR\_Analytics\_Dataset[Attrition] = "Yes"), COUNT(HR\_Analytics\_Dataset[EmpID]))**. Make it of % type and with one decimal digit.
- 

#### ☒ Dashboard Preparation Steps:

- ☒ Load canvas background image "HR\_Dashboard\_Background.jpg" with **Image fit:** Stretch
- ☒ For color code refer: <https://colorhunt.co/>
- ☒ Color code used here: **FFB4B4**  and **E66C37** 



- ☒ Insert **Text Box** with **Text:** HR ANALYTICS DASHBOARD, **Font:** Segoe (Bold), **Font size:** 20, **Bold:** Enabled, **Background:** No, **Color:** White, **Height:** 56, **Width:** 406
-  1. Insert **Card** with **Fields:** Count (Distinct) of EmpID, **Callout value (Color):** FFB4B4, **Category label:** No, **Height:** 97, **Width:** 175, **Title:** Count of Employees, **Text Color:** FFB4B4, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px
-  2. Insert **Card** with **Fields:** Sum of AttritionFlag, **Title:** Attrition
-  3. Insert **Card** with **Fields:** AttritionRate, **Title:** Attrition Rate
-  4. Insert **Card** with **Fields:** Average of Age, **Title:** Avg. Age, **Value decimal places:** 0
-  5. Insert **Card** with **Fields:** Average of MonthlyIncome, **Title:** Avg. Salary, **Value decimal places:** 1
-  6. Insert **Card** with **Fields:** Average of YearsAtCompany, **Title:** Years at Company, **Value decimal places:** 1



- ☒ Insert **Donut chart** with **Legend:** EducationField, **Value:** Sum of AttritionFlag, **Detail labels Values:** <Category, percentage of total>, Bold, FFB4B4, 0, **Height:** 263, **Width:** 354, **Title:** Attrition by Education, **Text Color:** FFB4B4, Center aligned, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px
- ☒ Insert **Stacked column chart** with **X-axis:** AgeGroup, **Y-axis:** Sum of AttritionFlag, **X-axis & Y-axis:**, Bold, FFB4B4, **Height:** 263, **Width:** 533, **Title:** Attrition by Age, **Text Color:** FFB4B4, Center aligned, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px





- ☒ Insert **Clustered bar chart** with **Y-axis:** SalarySlab, **X-axis:** Sum of AttritionFlag, **X-axis & Y-axis:**, Bold, FFB4B4, **Height:** 263, **Width:** 533, **Title:** Attrition by Salary Slab, **Text Color:** FFB4B4, Center aligned, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px
- ☒ Insert **Area chart** with **X-axis:** YearAtCompany, **Y-axis:** Sum of AttritionFlag, **X-axis & Y-axis:**, Bold, FFB4B4, **Lines color:** E66C37, **Height:** 284, **Width:** 532, **Title:** Attrition by Years at Company, **Text Color:** FFB4B4, Center aligned, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px, **Filters on this visual:** is less than 12



- ☒ Insert **Clustered bar chart** with **Y-axis:** JobRole, **X-axis:** Sum of AttritionFlag, **X-axis & Y-axis:**, Bold, FFB4B4, Adjust Y-axis width to 37%, **Height:** 284, **Width:** 362, **Title:** Attrition by Job Role, **Text Color:** FFB4B4, Center aligned, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px, **Filters on this visual:** JobRole, Top 5, By value Sum of AttritionFlag
- ☒ Insert **Matrix** with **Rows:** JobRole, **Columns:** JobSatisfaction, **Values:** Sum of AttritionFlag, **Values Text color and Alternate Text Color:** Bold, FFB4B4 and E66C37, **Height:** 259, **Width:** 362, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px



- ☒ Insert **Treemap** with **Category:** Gender, **Values:** Sum of AttritionFlag, **Colors:**, FFB4B4 and E66C37, **Height:** 142, **Width:** 185, **Title:** Attrition by Gender, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px
- ☒ Insert **Slicer** with **Field:** Department, **Colors:**, FFB4B4 and 6B2328, **Height:** 56, **Width:** 692, **Visual Border Color:** FFB4B4, **Rounded corners:** 13 px, **Width:** 3 px

